

Governance Under Load: Why Principal-Agent Systems Require Structural Governance, Not Relational Oversight

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Abstract

Principal-agent systems share a structural vulnerability: the principal is most likely to disengage from oversight precisely when the agent most needs guidance. This paper presents the Covenant Framework, a governance layer for multi-agent AI systems built on the premise that relational oversight degrades under load and structural governance must compensate. We ground the argument in three sources of evidence. First, the broader principal-agent and self-regulation literatures, supplemented by recent empirical work in organizational psychology showing that managers reduce feedback effort as perceived subordinate role ambiguity increases. Second, two production multi-agent sessions (AquaFlow, WedFlow) producing 9 agents across 3 generations with complete testament records and a documented synthesis case. Third, a Terminal-Bench 2.0 benchmark run in which a Covenant-governed agent scored 67.4 percent on 89 tasks (vanilla baseline 58.0 percent), and a controlled comparison in which Canon-derived rules outperformed ad-hoc prompt engineering by 40 percentage points on the same model. We formalize three propositions relating principal capacity, feedback quality, and system stability, then identify a structural asymmetry between human and AI principal-agent settings that has been underexplored: AI agents incur near-zero onboarding cost, which changes the governance design space in ways the human management literature does not anticipate. We close with limitations the benchmark adversary review surfaced, including an unresolved model confound, and outline the experimental program required to close them.

Keywords: principal-agent theory, AI governance, multi-agent systems, structural governance, runtime governance, distributed systems, agentic AI safety

1 Introduction

The manager and the AI orchestrator share an unlikely structural problem. Both are principals responsible for the performance and direction of agents operating beneath them (Jensen & Meckling, 1976; Eisenhardt, 1989; Bostrom, 2014). Both face finite resources of time, attention, and energy. And both are subject to a perverse dynamic: the conditions that most increase agents’ need for guidance are the same conditions that most constrain the principal’s ability to provide it (Lord et al., 2010; Schmidt & DeShon, 2007).

In the human organizational context, this dynamic now has direct empirical support. Recent longitudinal work surveying 352 managers found that managers reduced feedback effort as their perception of subordinate role ambiguity increased, the opposite of what self-regulatory theory had predicted (Vancouver & Day, 2005; Neal et al., 2017). The interpretation closest to the data is the goal-disengagement mechanism identified by Hollenbeck, Williams, and Klein (1989): when the discrepancy is perceived as too costly to close, in-

dividuals disengage entirely and reallocate effort elsewhere. Principals do not withdraw from governance because they are negligent (Bateman et al., 2002). They withdraw because the gap feels too wide to bridge.

This paper argues the same mechanism applies in multi-agent AI systems, and that the architectural response is the Covenant Framework: a governance layer that encodes policy structurally so it operates independent of principal availability. Figure 1 situates the framework relative to existing orchestration substrates.

Contributions. (1) We formalize the governance-under-load problem through three propositions linking principal capacity, feedback quality, and system stability. (2) We provide benchmark and session evidence that structural governance produces measurable performance differences, including a controlled comparison in which the same model and retry mechanism produced a 40-point performance gap depending on which rules governed the agent. (3) We identify a structural asymmetry between human and AI principal-agent systems that the human management literature does not antic-

ipate (Park et al., 2023; Wang et al., 2024): AI agents can be killed and respawned at near-zero cost, which widens the governance design space.

2 The Principal-Agent Problem Under Load

2.1 The Classical Problem and Its Hidden Assumption

Principal-agent theory (Jensen & Meckling, 1976; Eisenhardt, 1989; Akerlof, 1970) describes the delegation problem: a principal assigns tasks to an agent who has more local information and may pursue divergent interests. The canonical solutions are monitoring and incentive alignment (Eisenhardt, 1989). Both mechanisms quietly assume the principal is available to apply them. The literature has rarely interrogated this assumption directly (Bostrom, 2014; Dowding & Taylor, 2024), but it is exactly the assumption that breaks under load.

Self-regulatory theories of goal pursuit (Neal et al., 2017; Lord et al., 2010; Vancouver & Day, 2005) describe how individuals allocate finite resources across competing demands by the magnitude of perceived discrepancies between current and desired states. The classical reading is that larger discrepancies receive more resources (Schmidt & DeShon, 2007). The disengagement mechanism documented by Hollenbeck et al. (1989) complicates this picture: when a discrepancy is perceived as too costly to close, the rational response is not to allocate more resources but to abandon the goal and reallocate elsewhere. The agent who appears not to care is often the agent who has rationally concluded that caring will not produce the desired outcome within available constraints.

This mechanism is the bridge between organizational psychology and AI governance. In both contexts, the principal facing high load is not necessarily failing the agent through negligence (Bateman et al., 2002; Mintzberg, 1971). They are responding to a structural condition in which providing the needed governance is no longer perceived as achievable within available resources.

2.2 Empirical Anchor: Feedback Under Understaffing

Understaffing is defined as a condition in which workgroups have too few individuals or insufficient critical expertise to fulfill essential job functions (Hudson & Shen, 2015). Its consequences for front-line workers are well-documented: increased workload, elevated role ambiguity, and reduced well-being (Hudson & Shen, 2018; Shen et al., 2019; Gilboa, Shirom, Fried, & Cooper,

2008). What had been less well-studied until recently is the impact on managers, who are simultaneously the agents absorbing the consequences of understaffing and the principals expected to provide corrective guidance (Bateman et al., 2002; Miller, Waring, Bolton, & Sloane, 2019).

Recent empirical work provides the load-bearing anchor for this paper. In a four-wave longitudinal survey of 352 managers (Malo, 2024), understaffing was significantly positively related to both perceived subordinate role ambiguity ($b = .18, p < .001$) and managerial workload ($b = .26, p < .001$). However, the stage-two relationship inverted relative to prediction: perceived role ambiguity was negatively related to feedback effort ($b = -.29, p < .001$). Managers facing greater perceived ambiguity in their subordinates provided less feedback, not more.

The most theoretically significant interpretation is the goal-disengagement mechanism from Hollenbeck et al. (1989): when the gap feels too large to close, the rational response is to reallocate elsewhere. This is the structural condition we treat as motivating the entire design of the Covenant Framework, and it is the single empirical finding for which we rely directly on this thesis as a source. The broader theoretical apparatus is well-established in the principal-agent, self-regulation, and feedback literatures.

2.3 Feedback as a Clarity Mechanism

Feedback in the organizational context performs three structurally distinct functions. It reduces role ambiguity (Jong, 2016; Fisher & Gitelson, 1983), improves task performance (Kluger & DeNisi, 1996; Ashford & Tsui, 1991), and signals principal investment in the agent (van Dierendonck, Haynes, Borrill, & Stride, 2004; Steelman & Wolfeld, 2018). All three functions require the principal to be present, attentive, and willing (Linderbaum & Levy, 2010). None of them survive the disengagement mechanism intact.

The structural argument of this paper is that the function performed by feedback in human organizations is the same function performed by policy in AI multi-agent systems. Both communicate principal expectations, clarify scope, and constrain agent behavior toward desired outcomes. They differ in one critical respect: feedback is delivered relationally, in real time, dependent on the principal’s availability; policy is encoded structurally in advance and operates independent of principal availability. When load is low, the two are functionally interchangeable. When load is high, only the structural version still operates.

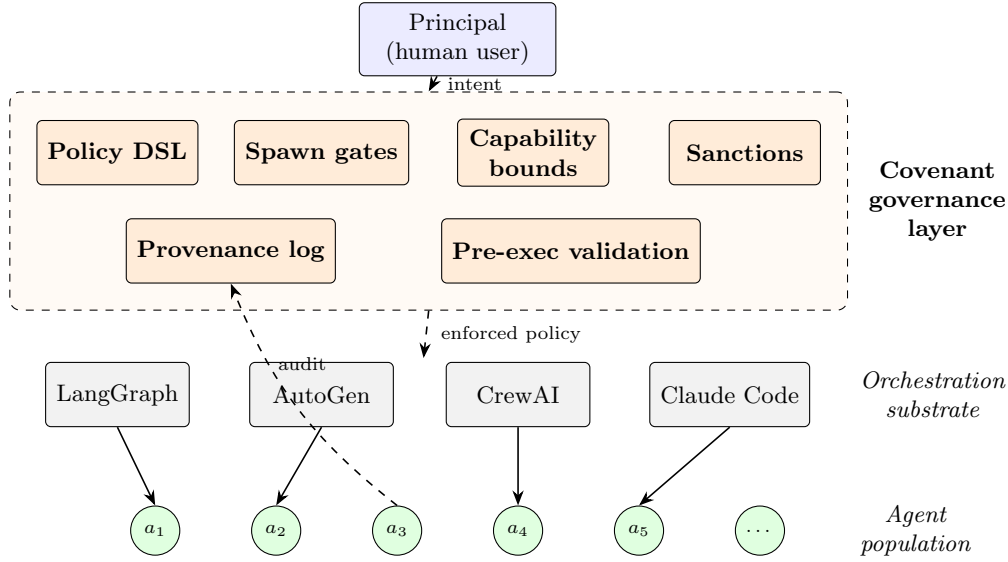


Figure 1: The Covenant Framework sits between the principal and the orchestration substrate. Existing frameworks (LangGraph, AutoGen, CrewAI, Claude Code) specify *what* agents do; Covenant specifies *what they are allowed to do*. The governance layer remains active when the principal is not, and emits an audit trail that survives the agent lifecycle.

3 The Covenant Framework

3.1 Motivation

The Covenant Framework (Salsali, 2025) is a governance layer for multi-agent AI systems. It was developed in response to a gap in contemporary agent orchestration. Frameworks such as LangGraph (LangChain, 2024), AutoGen (Wu et al., 2023), and CrewAI (Moura, 2024) specify the workflow of agent behavior, that is, what tasks are delegated, in what sequence, to which sub-agents. They provide minimal constitutional structure governing how agents may act, what resources they may consume, or how failures are categorized and contained (Raza et al., 2025; CMAG, 2026). Governance in these systems is either absent or implicitly delegated to a human operator monitoring in real time (Park et al., 2023; Wang et al., 2024).

This delegation breaks for the same reason the manager’s feedback effort breaks. In production environments with multiple concurrent agents operating across extended sessions, the principal cannot be present for every decision. The conditions that generate the greatest need for oversight, including high task volume, novel failure modes, cascading sub-agent spawning, and emergent inter-agent conflicts (van der Weij et al., 2024; Hubinger et al., 2024), are precisely the conditions that saturate principal attention. Relational governance fails when it is most needed. Structural governance does not. The conceptual lineage closest to the Covenant Framework is Constitutional AI (Bai et al., 2022), which similarly encodes behavioral expectations as natural-

language principles rather than relying on continuous human supervision, though our work operates at the level of multi-agent governance rather than single-model alignment.

3.2 Six Governance Primitives

The framework formalizes governance through six structural primitives.

Temporal policy enforcement encodes behavioral constraints as time-bounded rules in a domain-specific language. Policies define what agents may and may not do within specified contexts, eliminating the need for a principal to communicate expectations in real time.

Pre-execution validation checks agent action plans against policy before execution begins, catching violations proactively rather than reactively. This is structurally analogous to a manager reviewing a plan before approval, with the difference that it operates at every action without requiring manager availability and without the possibility of principal disengagement under load.

Capability bounding and quotas constrain the resources an agent may consume in a session. API calls, spawned sub-agents, and memory allocations are subject to per-identity budgets. This prevents resource exhaustion through unbounded delegation and ensures individual agent failures cannot cascade to system-level failure.

Graduated sanctions provide a tiered response to policy violations: warnings for minor deviations, capability restrictions for repeated violations, session ter-

mination for critical failures. This mirrors progressive discipline in human resource management, with the difference that it operates automatically and consistently, independent of principal cognitive state.

Provenance logging records agent actions, decisions, and justifications in a PROV-O-aligned audit trail. Multi-agent systems operating across sessions lose institutional context unless that context is externalized. Provenance logging is the framework’s mechanism for persistence and enables after-the-fact principal review and longitudinal policy refinement.

Spawn governance controls the conditions under which agents may create sub-agents, preventing unbounded delegation hierarchies and maintaining clear accountability chains throughout execution. Three pre-spawn checks must pass: overlap detection (no two agents working the same scope without coordination), scope validation (pause if total active agents exceed a threshold), and memory retrieval (search accumulated knowledge for prior work relevant to the proposed mandate before spawning).

3.3 The Central Design Choice

The unifying choice across all six primitives is that governance is encoded structurally rather than delivered relationally. The covenant document performs the function that feedback performs in the human management context: it communicates expectations, clarifies role boundaries, and enforces standards. The difference is that it does so without requiring a principal to be present, attentive, or willing. This is not a workaround for human absence. It is an architectural response to the empirically documented unreliability of relational oversight under load.

3.4 Mapping to Distributed Systems Concepts

The six primitives are framed in governance language because the paper’s central claim is a governance claim. Each primitive also has a direct analogue in the distributed systems literature, and we note these correspondences to ground the contribution in established systems vocabulary.

The mapping has two consequences for the paper. First, it positions Covenant as a layer that re-purposes well-understood distributed systems primitives for the LLM multi-agent setting, rather than as an invention from scratch. Second, it identifies where existing distributed systems theory can be imported. Admission control, rate limiting, and circuit breaker semantics are formally specified in operating systems and microservices literature (Nygard, 2018; Tanenbaum & Van Steen, 2017), and that formalism can be transferred to reason about Covenant’s behavior under load.

Table 1: Mapping of Covenant primitives to established distributed systems concepts.

Covenant primitive	Distributed systems analogue
Temporal policy enforcement	Declarative policy / constraint specification
Pre-execution validation	Static analysis / dry-run
Spawn gates	Admission control + backpressure
Capability bounds and quotas	Resource accounting / rate limiting
Graduated sanctions	Circuit breaker pattern
Provenance logging	Write-ahead log / audit trail

3.5 Operating Assumptions

The framework’s effectiveness depends on assumptions that this paper makes explicit so they can be tested independently.

A1. Policy interpretability.

Policies expressed in natural language are interpreted with low variance across runs by the underlying model. Reviewer feedback flagged this as the framework’s most consequential assumption; we return to it in Section 8.

A2. Good-faith reporting.

Agents produce testaments and provenance entries that accurately reflect their behavior. The Wed-Flow universal-success pattern (Section 4.2) is an empirical test of this assumption that the paper does not resolve.

A3. Capability threshold.

The underlying model has sufficient raw capability to follow the encoded rules. Below the threshold, governance produces no lift (Section 5.3).

A4. Deterministic gate evaluation.

Spawn-gate predicates are evaluated deterministically given identical inputs.

A5. Accurate resource accounting.

Token and tool-call counts in the manna ledger reflect actual consumption.

A6. Single coherent principal.

The framework currently models the principal as a single decision-maker; multi-principal extensions remain future work.

4 Results from Production Sessions

We report results from two sessions of the Covenant Framework on real codebases. These are descriptive and small-sample, and we are explicit about what they do and do not demonstrate. They show the framework operating end-to-end, producing the testaments and registry artifacts the architecture requires, and exercising the spawn gates and synthesis mechanism under realistic conditions. They do not constitute a controlled comparison against an ungoverned multi-agent baseline.

4.1 AquaFlow Session

The AquaFlow session, April 30, 2026, deployed the Covenant Framework on a water management application requiring both technical research and implementation output. Four agents were spawned across two generations: a Gen-1 Analyst researching the existing schema and notification infrastructure, a Gen-1 Writer building the weekly digest feature, and a Gen-2 Synthesist merging both parents' capability sets to add the corresponding UI control.

All four agents archived with complete testaments. The Analyst consumed approximately 33,000 tokens producing a brief identifying seven gaps and decisions for the Writer, including the existence of a missing `digest_enabled` column that required a database migration. The Writer reused two pure functions from the existing codebase without modification, diverged from the Analyst's recommendation in exactly one place (creating a separate `digest_log` table rather than altering the existing `notification_log` constraint), and documented the divergence in its testament. The Synthesist closed the gap between the new backend and the user-facing settings UI in a single agent lifecycle.

The Synthesist's testament is the most theoretically interesting artifact from the session. Synthesis under the Covenant Framework requires the child agent to report at sunset whether its emergent skill, that is, the capability inferred from combining two parents, proved real or theoretical. The Synthesist reported the skill was real and the synthesis was worth its cost, with the following evidence: the Analyst parent's database schema knowledge and the Writer parent's UI pattern knowledge were both required to set the correct default in the API destructuring and the state initialization. Neither parent alone could have completed the task in a single pass. The Analyst could find the schema gap but not build the UI. The Writer could build UI but had identified the gap without closing it. The synthesis eliminated a handoff that would otherwise have required a separate agent lifecycle and a distillation step.

The session passed all three spawn gates. Overlap detection found no conflict between the Analyst

and Writer mandates. Scope validation registered four agents, below the Babel threshold of six. Memory retrieval ran on each spawn and surfaced no prior work relevant to the mandates (this was the first session in this domain).

4.2 WedFlow Session

The WedFlow session, also April 30, 2026, deployed the framework on a wedding planning application requiring three concurrent mandate streams: an audit logging service, an escalation notification system, and three dashboard UX improvements (responsive mobile layout, fixing broken interactions, and replacing a placeholder guest list view with a real one). Six agents were spawned, all at Gen-1, with the root orchestrator providing coordination.

Five of the six agents archived with complete testaments (the sixth was a Gen-0 root that does not produce testaments). Total manna consumption across the five testament-producing agents was approximately 280,000 tokens, ranging from 31,000 (escalation-herald) to 82,000 (guests-activator). All five reported their mandates completed and all five recorded "Nothing failed" or equivalent on the `whatFailed` field, an outcome that surprised the operator and that we treat with appropriate skepticism in Section 8.

The session exercised the spawn gates under more concurrent load than AquaFlow. Five Gen-1 agents under one root brought total active count to six, exactly at the Babel threshold. The framework correctly registered this as the maximum acceptable concurrent agent count for a single session. Overlap detection caught a near-conflict between the `interaction-fixer` and `guests-activator` mandates, both of which initially scoped to "dashboard interactions" before the operator differentiated them (one to existing-component bug fixes, one to placeholder replacement).

The session illustrates how the framework behaves when mandates are largely independent rather than dependent. Unlike AquaFlow, where the Synthesist depended on outputs from both parent agents, WedFlow's mandates were parallel: an audit log and an escalation system are independent of the dashboard UX work. The framework's coordination overhead is correspondingly lower when independence allows parallelism.

4.3 What These Sessions Do Not Show

Neither session was run with an ungoverned control. We cannot say the framework produced better outcomes than an ungoverned multi-agent system on the same tasks. The session evidence demonstrates that the framework operates end-to-end, that the synthesis mechanism is non-trivial (the AquaFlow case has a clear emergent capability), and that the spawn gates catch

overlap conflicts in practice. The strongest comparative claim we can make from session data alone is that the framework is a working implementation, not a paper specification.

5 Benchmark Results: Terminal-Bench 2.0

The most rigorous comparative evidence in this paper comes from Terminal-Bench 2.0 (Tufano et al., 2025), a published benchmark of 89 terminal-based programming tasks scored by automated evaluators. Coding benchmarks for LLM agents have proliferated rapidly, with SWE-bench (Jimenez et al., 2024) and HumanEval (Chen et al., 2021) representing earlier methodological touchstones, though both have known validity issues including solution leakage and weak test cases (Yu et al., 2024; Aleithan et al., 2024). Terminal-Bench tests multi-turn terminal workflows including compiling, debugging, and environment setup, providing a more agent-realistic evaluation than function-level benchmarks. We report a full-benchmark score, a small-sample controlled comparison that isolates the contribution of governance rules from the underlying model, and an unresolved model confound that we are explicit about.

5.1 Full Benchmark Run

The full Terminal-Bench 2.0 run, completed May 6, 2026, scored 67.4 percent across all 89 tasks with the Covenant Prophet v3 agent using Claude Opus 4.7, single attempt, no retry, no timeout multipliers, no resource overrides. The published vanilla Claude Code baseline using Claude Opus 4.6 scored 58.0 percent. This places the governed agent approximately at rank 10 on a leaderboard topped by frontier-lab agent scaffolds at 79 to 82 percent (Codex CLI with GPT-5.5, ForgeCode with GPT-5.4, TongAgents with Gemini 3.1 Pro, ForgeCode with Claude Opus 4.6).

The Covenant Prophet v3 agent is a Harbor-compatible adapter that wraps Claude Code CLI with a six-rule governance system prompt, a five-step workflow definition, and a testament retry mechanism that extracts failure context and retries once with that context prepended. The six rules are distilled directly from the full Canon: read environment before coding, plan before executing, diagnose errors before retrying, verify before declaring done, conserve context tokens, and step back after three failures. The full distilled prompt is approximately 800 bytes.

5.2 E2: Canon-Derived Rules vs Ad-Hoc Rules

The most controlled comparison in the benchmark run held everything constant except the rules. Same model (Opus 4.7), same architecture (Harbor adapter), same retry mechanism, same 10 tasks, different rules.

The ad-hoc rules were generated by asking an LLM to "write the best 6 rules for a terminal benchmark agent" with no access to the Canon. They are competent rules of the kind a careful prompt engineer would write: read the task carefully, diagnose errors rather than retry blindly, write small testable increments, step back after two failures, prefer simple languages, run all provided tests before declaring completion.

The two rule sets overlap substantially. Both require reading before coding, both require error diagnosis, both require verification. The differences are subtle. The Canon-derived rules are more behavioral (what to do when stuck, how to manage scarce context tokens) while the ad-hoc rules are more procedural (write small increments, use simple languages). The behavioral framing produced double the success rate on the same tasks with the same model.

We treat this as the cleanest single-experiment evidence in the paper. It is same-model controlled, same-task controlled, and architecturally interesting because it challenges the assumption that more elaborate governance produces better agents. A 32KB version of the full Canon, tested in an earlier iteration, scored 60 percent on the same 10 tasks: the slimmer rule set outperformed the heavier one by 20 points, presumably because the heavy Canon consumed context tokens the model needed for reasoning.

Statistical note. A Fisher’s exact test on the underlying 8/2 vs 4/6 contingency yields $p \approx 0.17$, which does not clear conventional significance thresholds at $n = 10$. We treat the result as directionally consistent and underpowered, and Section 8 identifies a 40–60 task replication as the necessary next step.

5.3 Capability Threshold

A separate experiment varied the underlying model while holding governance constant. Three Anthropic models were tested with the Covenant Adaptive v5.0 agent on the same 10 tasks: Opus 4.7, Sonnet 4.5, and Haiku 4.5.

The Haiku result is the theoretically interesting one. Haiku scored zero on a clean run with no infrastructure issues, indicating a minimum capability threshold below which governance is irrelevant. The model lacks the raw coding ability to solve the tasks regardless of how disciplined its execution is. Above this threshold,

Table 2: Summary of the two production sessions. Manna is approximate token consumption per agent. Mandate dependency indicates whether downstream agents required outputs from upstream agents.

	AquaFlow	WedFlow
Date	2026-04-30	2026-04-30
Agents spawned	4	6
Generations	2	1
Archived with testaments	3 of 3 child agents	5 of 5 child agents
Total manna	~66k tokens	~280k tokens
Manna range per agent	33k tokens (Analyst)	31k–82k tokens
Synthesis events	1 (emergent skill: real)	0
Spawn-gate triggers	0	1 (overlap caught)
Active-count peak	4	6 (at Babel threshold)
Mandate dependency	Sequential	Parallel

Table 3: Terminal-Bench 2.0 leaderboard context. The Covenant Prophet result is positioned between the frontier-lab agent scaffolds (which use proprietary multi-step planning frameworks) and the vanilla Claude Code baseline.

Rank	Agent	Underlying model	Accuracy
1	Codex CLI	GPT-5.5	82.0%
2	ForgeCode	GPT-5.4	81.8%
3	TongAgents	Gemini 3.1 Pro	80.2%
4	ForgeCode	Claude Opus 4.6	79.8%
8	Capy	Claude Opus 4.6	75.3%
~10	Covenant Prophet	Claude Opus 4.7	67.4%
40	Claude Code (vanilla)	Claude Opus 4.6	58.0%

Table 4: E2 controlled comparison. Same model (Opus 4.7), same architecture, same retry mechanism, same 10 tasks. Only the rules differed.

Agent	Rules source	Score	Delta
Covenant Prophet v3	Canon-derived (6 rules)	80%	+40
Ad-hoc base-line	Prompt-engineering intuition	40%	—

governance produces measurable lift. At Opus capability, the lift is 12 to 22 points depending on the baseline used. At Sonnet capability, the lift is from near-zero to 20 percent, a smaller absolute gain but proportionally larger. The implication for downstream work is that a mid-tier model fine-tuned on governance patterns may approach frontier-model performance on agentic tasks at a fraction of the cost.

5.4 What the Benchmark Does Not Resolve

An internal adversary review surfaced an unresolved model confound that we acknowledge directly. The 67.4 percent governed result uses Opus 4.7, while the

published vanilla baseline of 58.0 percent uses Opus 4.6. Some portion of the nine-point delta could be attributable to the model upgrade rather than the governance layer. Resolving this requires running vanilla Opus 4.7 (no governance, no system prompt) on the same 89 tasks. That experiment is planned but not yet completed at time of writing.

The E2 result (Canon-derived 80 percent vs ad-hoc 40 percent) is not subject to this confound because both arms use Opus 4.7 with the same retry mechanism, but the sample is 10 tasks rather than 89. Both pieces of evidence point in the same direction, and the E2 result is methodologically cleaner. The full-benchmark result is methodologically less clean but has a larger sample. We let the reader weight these tradeoffs rather than papering over them.

The full-run sample of 89 tasks is also enough to project leaderboard placement but not enough to test whether governance effects vary across task categories (algorithmic, system-level, debugging-heavy). We expect category effects exist; characterizing them is a planned next experiment.

Table 5: Capability threshold experiment. Same governance prompt (Covenant Adaptive v5.0), same 10 tasks, different underlying models. Single-run results; replication with mean $\pm$ SD is a planned next experiment (Section 8).

Model	Score	Interpretation
Claude Opus 4.7	70–80%	Governance lifts above baseline by 12–22 points
Claude Sonnet 4.5	20%	Governance lifts a marginal model above zero
Claude Haiku 4.5	0%	Below capability threshold; governance cannot compensate

6 Three Propositions

The principal-agent framing, the production session evidence, and the benchmark results support three formal claims about governance under load.

Proposition 1 (Feedback–Policy Equivalence). High-quality managerial feedback and structurally encoded policy are functionally equivalent mechanisms for communicating principal expectations to agents, reducing scope ambiguity, and supporting agent performance. They differ in their dependence on principal availability. Feedback degrades as principal load increases. Policy does not.

The empirical support comes from two directions. In the human context, feedback’s role as a clarity mechanism is well-documented (Jong, 2016; Kluger & DeNisi, 1996; Ashford & Tsui, 1991; Fisher & Gitelson, 1983), as is its degradation under principal load (Hollenbeck et al., 1989; Schmidt & DeShon, 2007; Lord et al., 2010). In the AI context, the E2 result demonstrates that the specific content of policy matters: Canon-derived rules produced double the success rate of competent ad-hoc rules on the same model and tasks. This parallels findings in Constitutional AI showing that natural-language principles produce measurable behavioral shifts (Bai et al., 2022). The two contexts converge on the same underlying claim: principal expectations are mechanically transmitted to agents through structured communication, and the quality of that communication is causally connected to agent performance.

Proposition 2 (Principal Disengagement Under Load). Under conditions of high principal load, relational governance mechanisms degrade monotonically with the magnitude of the governance gap. Structural governance mechanisms are invariant to principal load within their operational parameters. Principal-agent systems operating in high-load environments therefore require structural governance as a necessary component of their design, not as a supplement.

The empirical support for the relational degrada-

tion claim comes from the goal-disengagement literature (Hollenbeck et al., 1989; Schmidt & DeShon, 2007; Neal et al., 2017) and recent confirmation in the management context, where managers facing high subordinate ambiguity reduced rather than increased feedback effort. The empirical support for the invariance claim comes from the benchmark architecture: the governed agent’s six rules apply identically to every task without regard to operator availability, a property that parallels constitutional approaches to LLM alignment (Bai et al., 2022; Raza et al., 2025), and the testament logging continues even when no one is reading it. The benchmark could not have run for 90 hours on a single API session with continuous principal attention. It ran because the governance layer did not require it.

Proposition 3 (Agency–Governance Complementarity). Within principal-agent systems, governance specificity and agent autonomy are complements rather than substitutes. Structural governance that clearly encodes capability bounds, sanctioned behaviors, and escalation conditions enables greater productive autonomy than underspecified relational oversight, because it eliminates the ambiguity that constrains agent action.

The cleanest evidence for this proposition is the AquaFlow synthesis case. The Synthesist’s testament explicitly attributes its single-pass success to the precision of the mandate: the parent agents had identified the exact file, the exact location within the file, and the exact pattern to match, which “made the implementation a pure execution task.” Underspecified relational oversight would have required the Synthesist to ask clarifying questions or guess. Well-specified structural governance allowed it to act decisively within known bounds. This is the inverse of the conventional governance-versus-autonomy tradeoff: the right governance structure does not restrict agency, it constitutes the conditions under which meaningful agency is possible.

7 The Human-AI Lifecycle Asymmetry

The principal-agent framing is productive precisely because it identifies a shared structural problem across two superficially different domains. It is less productive when it papers over genuine asymmetries between those domains. The most consequential asymmetry, raised in early review of this work, concerns the cost of agent turnover (Figure 2).

In the human management context, hiring and firing carry substantial costs. Recruitment, onboarding, knowledge transfer, and team disruption are real expenses. A manager dealing with an underperforming employee cannot trivially replace them; the cost of replacement is often higher than the cost of continued underperformance. This is one reason the goal-disengagement mechanism is so consequential in the human context: the manager who concludes that closing the gap is too expensive does not have a cheap alternative path.

In the AI context this constraint largely vanishes. An agent that produces a poor testament can be archived and respawned with refined context at the cost of a few thousand tokens. There is no team disruption (the other agents do not know the agent was respawned), no onboarding (the testament from the previous attempt becomes inheritance for the next one), and no knowledge transfer cost (the relevant knowledge is externalized in provenance logs). The expensive thing in human management, that is, replacing the agent, is the cheap thing in AI multi-agent systems.

This asymmetry reshapes the governance design space in three concrete ways.

First, it weakens the case for graduated trust models in AI contexts. The orientation primitive gap, raised in earlier discussions of this framework, was framed by analogy to a probationary period in human resource management: new agents would begin with tighter capability bounds that relax as compliance accumulates. The human analogue exists because firing a probationary employee is expensive enough that gradual trust is the dominant strategy. For AI agents, an alternative strategy is available: instead of relaxing bounds over time, just archive and respawn the agent with cleaner context when it deviates. The orientation primitive remains useful for tasks with state that survives across agent lifetimes, but it is no longer obviously the only architectural response.

Second, it strengthens the case for aggressive sanctions. In the human context, graduated sanctions exist partly because the alternative (termination) is costly. In the AI context, session termination of an underperforming agent costs almost nothing, which means the framework can afford to terminate sooner and with

less hesitation than human management would tolerate. The graduated sanction structure in the Covenant Framework currently mirrors the human progressive discipline model; an AI-native sanction structure might be steeper.

Third, it complicates the validity of the original finding as a one-to-one analogy. The manager who reduces feedback effort under load is responding partly to the fact that the underperforming subordinate cannot be cheaply replaced. The AI principal facing analogous load has an option the manager does not: terminate and respawn. Whether the goal-disengagement mechanism applies with the same force in the AI context is an empirical question that has not been studied. The structural claim of this paper, that relational governance degrades under load, survives the asymmetry. The specific mechanism may differ in ways that change the architectural response.

We treat this asymmetry as a feature of the AI governance design space, not a refutation of the cross-domain framing. The shared structural problem is real, and the architectural response (encoding governance structurally rather than relying on principal availability) holds in both domains. The difference is that the AI version has more degrees of freedom, and the framework can take advantage of them.

8 Critical Gaps and Limitations

We close with the open questions and limitations that the work has not resolved. This section is informed by an internal adversary review of the benchmark findings, and we report its conclusions directly rather than minimizing them.

Model confound. The 67.4 percent governed full-benchmark result uses Opus 4.7. The 58.0 percent published vanilla baseline uses Opus 4.6. Some portion of the nine-point delta could be attributable to the model upgrade rather than the governance layer. The planned mitigation is a vanilla Opus 4.7 run on the same 89 tasks. The E2 result (80% vs 40%, both Opus 4.7) is not subject to this confound but uses a smaller sample of 10 tasks.

Retry contribution. No benchmark run isolates governance rules alone from governance rules plus retry. The testament retry mechanism converts an estimated 10 to 15 percent of failures to passes, but its independent contribution cannot be precisely separated from the rules' contribution without a controlled experiment. The mitigation is a run with retry disabled on the same task set.

Session sample size. Two production sessions with nine combined archived agents is descriptive evidence at best. The patterns observed (testament completion, spawn gate function, synthesis emergent capability) are

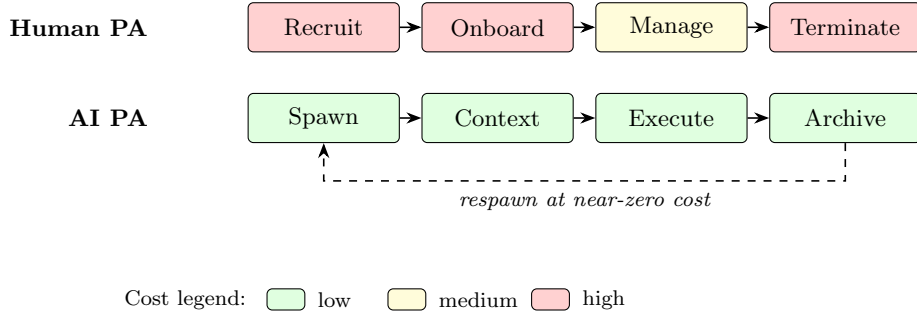


Figure 2: Lifecycle cost asymmetry between human and AI principal-agent (PA) systems. In human PA, recruit and terminate are the dominant cost drivers, constraining the principal’s response options under load and producing the goal-disengagement mechanism documented in the management literature. In AI PA, every stage is low-cost and the dashed loop represents an option unavailable in human management: when an agent underperforms, archive it and respawn with refined context. This degree of freedom is the basis for Section 7’s claim that AI governance design has more degrees of freedom than the human analogue suggests.

consistent with the framework operating as designed, but the sample does not support statistical claims about outcomes.

Statistical power on E2. The 80% vs 40% comparison on 10 tasks is suggestive but underpowered. A Fisher’s exact test on the 8/2 vs 4/6 contingency yields $p \approx 0.17$, which does not clear conventional significance thresholds. The directional consistency of the result and the same-model, same-architecture control are the reasons it is treated as the cleanest evidence in the paper, but a larger sample (estimated 40–60 tasks) is needed to make the effect statistically defensible.

Policy interpretation variance (A1). The framework assumes natural-language policies are interpreted with low variance across runs by the underlying model. In practice, LLM agents perform fuzzy interpretation: a rule like “step back after three failures” requires the agent to define what constitutes a failure. Stochastic decoding, context length, and model version all influence interpretation. Reviewer feedback flagged this as the framework’s most consequential unstated assumption. The mitigation is empirical: measure inter-run variance on a fixed policy and report it as a property of the governance layer.

Universal success in WedFlow. All five testament-producing agents in WedFlow reported “Nothing failed” on the `whatFailed` field. This is the field most vulnerable to motivated reporting from agents that may not want to surface their own failures, a concern that parallels emerging evidence of strategic underperformance and deception in evaluation contexts (van der Weij et al., 2024; Hubinger et al., 2024). The framework’s auditor agent (Guardian) reads testaments against actual manna logs and registry state, but the audit was not run on WedFlow before this paper. Uniform “nothing failed” reports should be treated as a flag for follow-up audit, not as evidence of universal success.

Single runtime. All sessions and benchmark runs use Claude Code with Claude models. The compliance checklist and the six distilled rules may contain implicit assumptions about Claude’s capabilities that do not transfer to other underlying models. Cross-model validation is planned but not completed.

Goal-disengagement mechanism in AI principals. The finding that managers withdraw feedback under perceived role ambiguity maps onto the human context cleanly. Whether the same mechanism applies with the same force when the principal is itself an AI orchestrator, or when the cost of agent replacement is near-zero (Section 7), is an open empirical question. The structural claim survives the asymmetry. The specific mechanism may not.

Absence of game-theoretic formalization. The paper develops three propositions in natural language but does not provide a formal game-theoretic treatment of the principal-agent interaction under load. A proper formalization would model the interaction as a sequential game with information asymmetry and verify the propositions analytically rather than empirically. This is acknowledged as a substantial future work item rather than a gap closable within this paper.

9 Conclusion

The manager who cannot find time to give feedback and the orchestration framework that lacks a governance layer are expressions of the same structural failure: the assumption that relational oversight will be available when it is most needed. The empirical evidence from organizational psychology says this assumption fails systematically under load, not because principals are negligent, but because the structural conditions that create the need for governance are identical to those that consume the principal’s capacity to provide it. The archi-

tectural response is to encode governance in policy that operates independent of principal availability.

The Covenant Framework operationalizes this response through six governance primitives. The benchmark and session evidence support the claim that structural governance produces measurable performance differences: a 40-point gap between Canon-derived and ad-hoc rules on the same model and tasks, a 9.4-point lift on the full Terminal-Bench (subject to an acknowledged model confound), and a successful synthesis case in production that closed a handoff a sequential pipeline would have required. The three propositions formalize the underlying claims. The human-AI lifecycle asymmetry identifies where the cross-domain framing requires care.

What we have not done, and what the work needs next, is a controlled comparison against ungoverned multi-agent baselines on the same production tasks, a vanilla Opus 4.7 run to resolve the model confound, and characterization of governance effects across task categories. The benchmark adversary review surfaced these gaps clearly, and we report them rather than minimize them. The contribution of this paper is the framing, the propositions, the architecture, and the first round of evidence. The next round of evidence is what closes the gap between an argument and a robust empirical claim.

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